Hemodynamic Digital Twins

A Physics-Informed Artificial Intelligence Approach for Critical Care Prognostics

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PROJECT CONTEXT & MOTIVATION

In intensive care units (ICUs), hypotensive shock can cause irreversible organ failure within minutes. Despite monitoring patients at 100Hz (100 samples per second), standard medical alerts are strictly reactive, triggering only after catastrophic deterioration has occurred. Furthermore, applying traditional deep learning (like Transformers) to this high-frequency data results in a quadratic memory bottleneck $\mathcal{O}(L^2)$, instantly crashing standard hospital hardware (the "Needle in a Haystack" limitation).

CORE OBJECTIVE

This thesis proposes the construction of a proactive **Medical Digital Twin**, predicting Mean Arterial Blood Pressure (MBP) up to 30 minutes into the future. By hybridizing lightning-fast sequence models with pure fluid dynamics, the system achieves zero-latency bedside inference while maintaining clinical interpretability.

ARCHITECTURAL CONTRIBUTIONS

The project successfully engineers three distinct, novel neural architectures to overcome both computational and biological limitations:

- **Medical_MambaTS (Edge Tier):** A hardware-aware state space model (SSM) that achieves linear-time $\mathcal{O}(L)$ scaling. To combat messy ICU data, we engineered a *Missingness-Aware Gate* that injects $-\infty$ when a sensor drops, forcing the state transition to the Identity matrix (I) to freeze the memory and prevent biological corruption.
- **SST-KODA (Cloud Tier):** A Spectral State-Space Koopman Operator. It utilizes a gating mechanism to isolate high-frequency noise from low-frequency organ trends, lifting the non-stationary, non-linear biological data into an infinite-dimensional linear space for highly accurate forecasting.
- **PI-DeepONet (Physics Tier):** Traditional Physics-Informed Neural Networks (PINNs) require hours of backpropagation to calibrate to a new patient. PI-DeepONet solves this via Operator Learning. A *Branch Network* ingests vital signs to extract patient-specific parameters (Vascular Resistance & Compliance), while a *Trunk Network* tracks continuous time. Constrained by a soft-penalty 0D Windkessel ODE ($\mathcal{Q} = C \cdot dP/dt + P/R$), it calibrates to a new patient in **0.20 milliseconds** without retraining.

DATA INTEGRITY & BENCHMARKING

The models were trained on the high-fidelity **VitalDB** dataset. To absolutely prevent data leakage, scaling and standardization (μ, σ) were calculated strictly on a per-patient basis. A rigorous 70/15/15 patient-level split ensured that testing data remained a true "blind future." Against robust baselines (Informer, LSTMs), the hybrid approach drastically reduced Mean Squared Error (MSE) by severely penalizing sudden hypotensive drops over safe averages.

EXPLAINABLE AI (XAI) & DEPLOYMENT

Because physicians cannot administer lethal vasopressors based on black-box alerts, the system utilizes a multi-modal XAI strategy. We employed **Integrated Gradients** and **Self-Attention Maps** to highlight temporal landmarks (like the dicrotic notch), while the PI-DeepONet explicitly outputs causal physical parameters (e.g., collapsing vascular resistance).

The entire framework is deployed as a real-time, interactive **Next.js / React Three Fiber 3D Dashboard**, backed by a **FastAPI** engine utilizing dynamic GPU lazy-loading to prevent VRAM overflow.

CONCLUSION

This research proves that a hybrid Edge-Cloud architecture embedded with soft-penalty fluid dynamics is the definitive solution for critical care prognostics, achieving both state-of-the-art computational latency and indispensable causal transparency.